

TECHNICAL UNIVERSITY OF MOLDOVA
Faculty of Computers, Informatics, and Microelectronics
Department of Software Engineering
Software Engineering Study Programme

LABORATORY REPORT

OPERATING SYSTEMS LABORATORY 4

The xv6 Shell from the Outside

Student: Lucian-Adrian Gavril

Group: FAF-241

Coordinator: univ. assist., Reițman Patricia

Chișinău, 2026

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ABSTRACT

This report examines how the xv6 shell turns command text into process execution. It explains file redirection, pipelines, command sequencing, the shell command tree in `user/sh.c`, and why the built-in `cd` must run in the parent process. It also inspects the `sleep` and `args` utilities and compares their argument handling with GNU Bash. A 9 October 2026 QEMU run verified the no-argument `sleep` message and `sleep 20`, which returned to the prompt in 1.96 seconds. A 10 October QEMU run verified all four requested `args` cases. The screenshots are genuine terminal captures from Dev1 Debian GNU/Linux 13; they do not establish that a course-assigned laptop or VM was used.

Keywords: operating systems, xv6, shell architecture, command line, process hierarchy, argument passing, riscv

INTRODUCTION

Operating system shells mediate the boundary between user input and kernel process services. A shell parses commands, configures file descriptors, starts processes, creates pipelines, and invokes `exec()` [1, 2].

Laboratory 4 examines the xv6 shell (`user/sh.c`) alongside GNU Bash to identify core shell primitives and the user-space features omitted by xv6. The host did not have a globally installed RISC-V toolchain or QEMU, and `sudo -n true` returned exit code 1 with `sudo: a password is required`. Official Debian packages were therefore extracted under a temporary `/tmp` prefix without installing host packages. On 9 October, `make qemu` booted a temporary copy of the course tree. A second focused `make qemu` run on 10 October tested all four argument patterns. The figures below show genuine Bash and QEMU captures from the Dev1 Debian GNU/Linux 13 host; they do not establish that the course-assigned laptop or VM was used.

The investigation is divided into four parts:

a) **System Call Verification and Sleep Implementation**: Validating the kernel system call declaration in `user/user.h`, confirming proper argument parsing and tick delays in `user/sleep.c`, and registering the utility in the xv6 build system;

b) **Comparative Behavioral Analysis**: Testing identical syntax patterns in Bash and xv6 across I/O redirection, pipelines, list grouping, sequential execution, background jobs, globbing, environment variables, quotation semantics, path searching, and command-line editing;

c) **Argument Vector Inspection**: Authoring dual implementations of an argument printer (`args.c` for xv6 and `args_linux.c` for Linux) to determine who expands wildcards and strips quotation marks;

d) **Source Code Architecture of `user/sh.c`**: Examining the tagged command structures, recursive execution in `runcmd()`, and the explicit parent-process handling of the `cd` command;

1 SYSTEM CALL VERIFICATION AND SLEEP IMPLEMENTATION

1.1 System call declaration in user.h

In xv6, system calls available to user programs are prototyped in `user/user.h`. Because system calls represent kernel traps triggered by assembly stubs in `usys.S`, the function signatures in `user.h` must match the dispatch tables in `kernel/syscall.c`.

The source capture shows the system-call declaration in the copied header (Figure 1.1) [1].

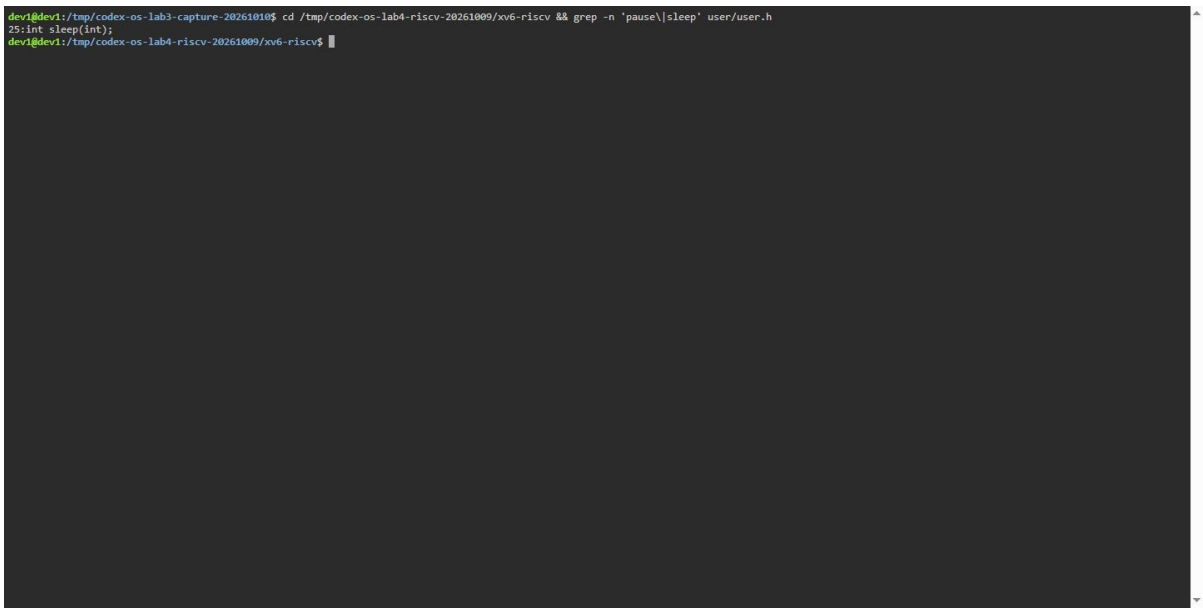


Figure 1.1 – Terminal capture, Dev1 Debian GNU/Linux 13, 10 October 2026: System call declaration verified within user/user.h.

1.2 Implementation of user/sleep.c

The `sleep` utility serves as an interface between command-line arguments and the kernel timer tick scheduler. The utility verifies that exactly one argument is supplied (`argc == 2`). If the parameter count is invalid, an error line is emitted to file descriptor 2 (stderr), and the process exits with status 1. If valid, `atoi()` parses the argument string into an integer and passes it to the system call (Figure 1.2).

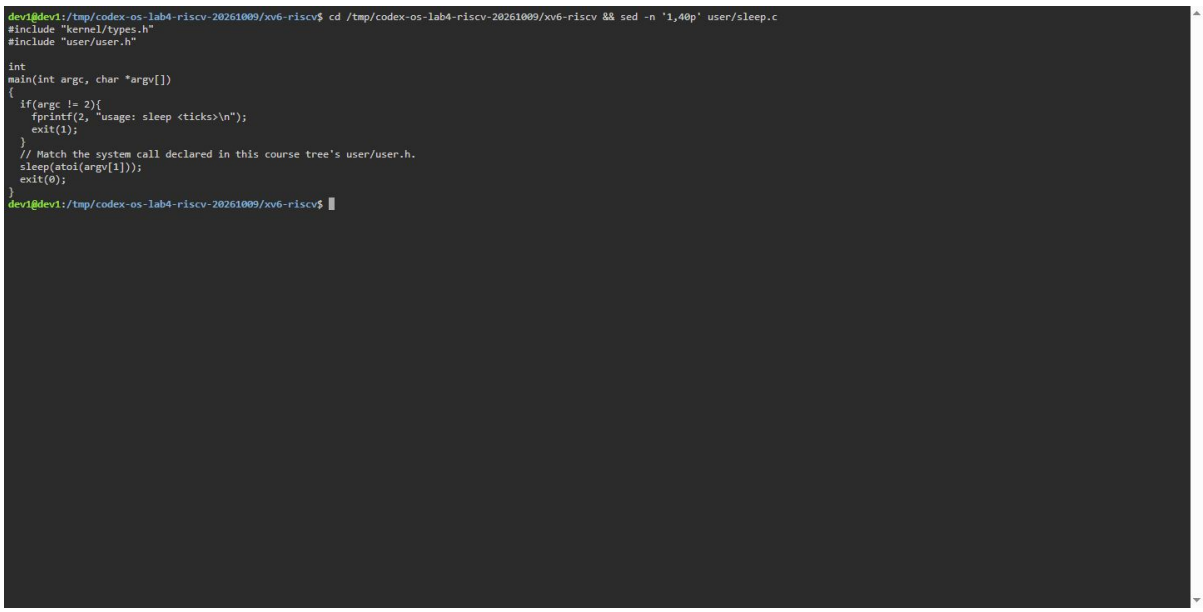


Figure 1.2 – Terminal capture, Dev1 Debian GNU/Linux 13, 10 October 2026: Implementation of user/sleep.c with input validation.

1.3 Runtime execution in xv6

The 9 October QEMU run returned to the prompt after `sleep 20` in 1.96 seconds of measured wall time. It also printed the no-argument usage message and checked two `args` cases. The following excerpt preserves that earlier serial transcript; the 10 October four-case run is shown in Figure 3.3:

```
init: starting sh
$ sleep
usage: sleep <ticks>
$ sleep 20
$ args one two three
argc = 4
argv[0] = [args]
argv[1] = [one]
argv[2] = [two]
argv[3] = [three]
$ args *.c "x y"
argc = 4
argv[0] = [args]
argv[1] = [*.c]
argv[2] = ["x]
argv[3] = [y"]
$ echo LAB4_SLEEP_DONE
LAB4_SLEEP_DONE
$
```

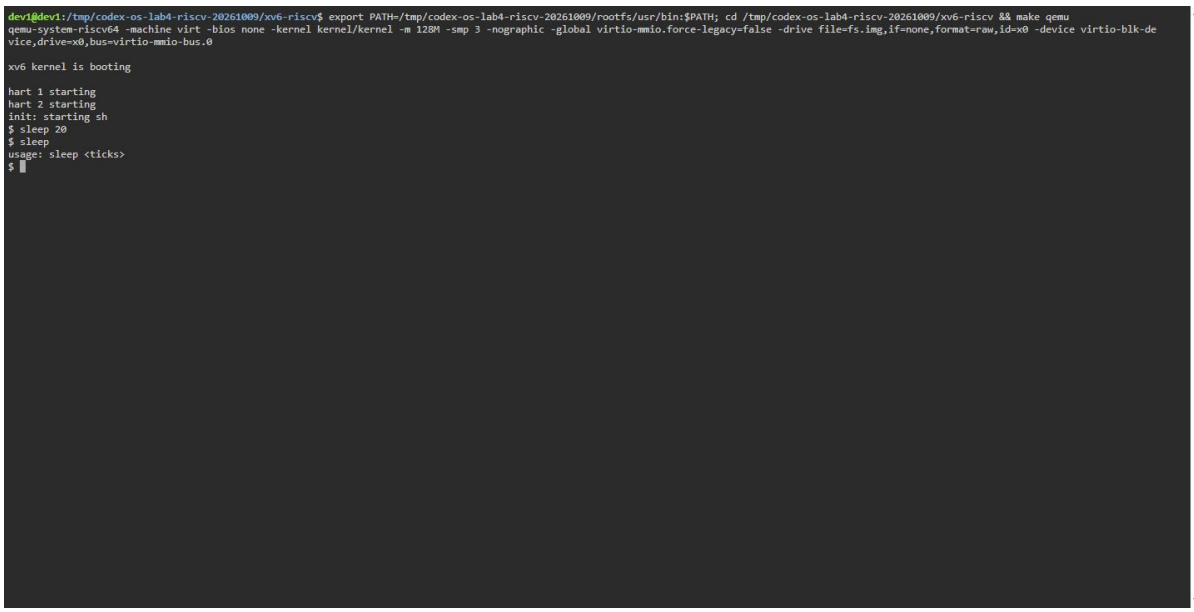


Figure 1.3 – Terminal capture, Dev1 Debian GNU/Linux 13, 10 October 2026: xv6 boot, `sleep 20`, and the no-argument usage message.

2 COMPARATIVE ARCHITECTURE: BASH VERSUS XV6 SHELL

2.1 Systematic comparison matrix

The matrix records observed output for nine shell commands and separate keyboard-behavior checks. The Bash and xv6 terminal captures show command rows 1–9; Bash history, completion, and reverse-search behavior and the xv6 line-editing behavior were checked separately. Explanations are based on the course tree and shell source [1, 3, 2].

Test Paradigm	Bash Output	xv6 Output	Architectural Explanation
Redirection `echo hello > f ; cat < f`	`hello`	`hello`	Both implement `open()` and `dup()` redirection via file descriptors.
Pipe `ls | grep READ`	`README`	`README`	Both create kernel pipe buffers and coordinate reader/writer via `pipe()`.
Grouping `(echo a; echo b) | wc`	`2 2 4`	`2 2 4`	Both shells accept the tested grouped command in a pipeline; `wc` reports lines, words, and bytes.
Sequence `echo one ; echo two`	`one` `two`	`one` `two`	Both implement sequential command execution via `LIST` nodes separated by `;`.
Background `echo bg &`	`bg` (background)	`bg`	Both support asynchronous child execution by omitting `wait()`.
Globbing `ls *`	Lists all entries	`ls: cannot open *`	Bash performs pathname expansion; xv6 passes literal wildcard characters.
Variables `echo $HOME`	`/home/dev1`	`$HOME`	Bash expands the host variable; xv6 prints the token literally.
Quotes `echo "a b"`	`a b` (single arg)	`"a b"` (literal)	Bash removes quotes and protects whitespace; xv6 splits strictly on spaces.
PATH Lookup `mkdir d ; cd d ; ls ..`	Lists parent-directory entries	`exec cd failed`; root entries listed	Bash runs `cd` in the list and uses `$PATH`; xv6 tries `exec cd`, then runs `ls ..` from `/`.
Line Editing `Arrows, Tab, ^R`	Up: history; Tab: completion; Ctrl+R: reverse search	Up/Ctrl+R: no history or search; blank-prompt Tab: no completion list	Bash behavior was checked interactively. The focused QEMU check found no Up-arrow recall or Ctrl+R search. Blank-prompt Tab produced no completion list; path-prefix completion was not tested. Source inspection shows no such handler in the xv6 console.

Table 2.1 – Command results for rows 1–9; line-editing behavior was checked separately in Bash and QEMU.

2.2 Shell command captures

The Bash capture shows the nine shell commands run on Dev1 Debian GNU/Linux 13. Up-arrow, Tab, and Ctrl+R behavior was checked separately in an interactive Bash session.

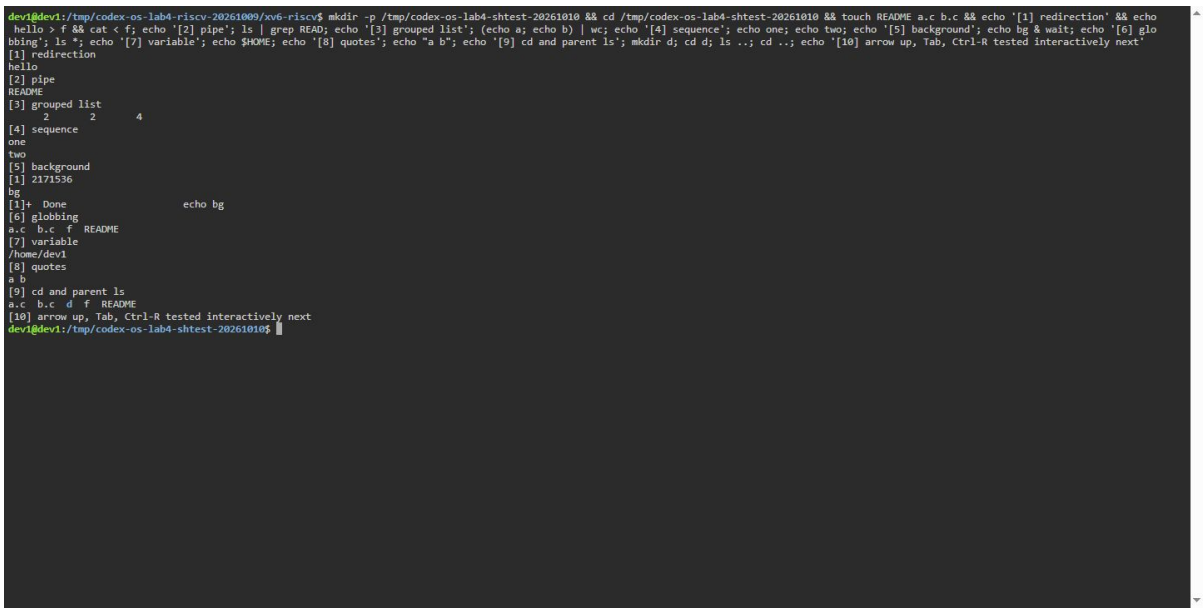


Figure 2.1 – Terminal capture, Dev1 Debian GNU/Linux 13, 10 October 2026: Bash results for command rows 1 through 9.

The xv6 capture shows nine shell commands. An interaction tested Up, Tab, and Ctrl+R in QEMU: Up and Ctrl+R showed no history or search; Tab at an empty prompt produced no completion list. The console source in `kernel/console.c`, lines 140–177, handles Ctrl+P, Ctrl+U, Ctrl+H, and Delete. Path-prefix completion was not tested.

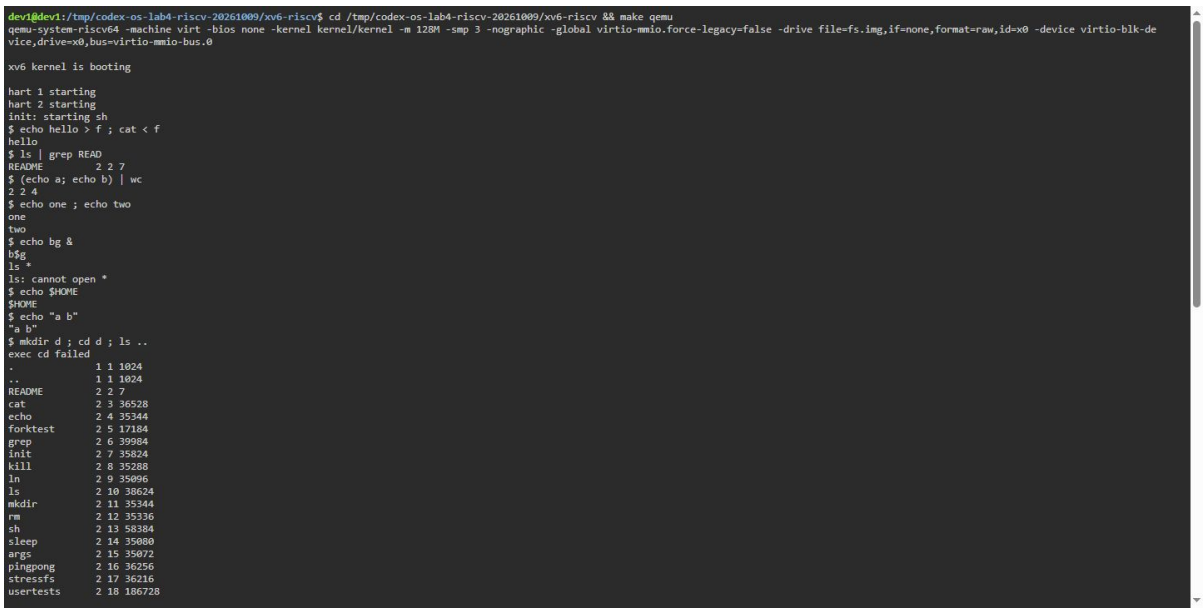


Figure 2.2 – Terminal capture, Dev1 Debian GNU/Linux 13, 10 October 2026: xv6 QEMU results for comparison rows 1 through 9.

2.3 Path resolution and directory navigation

The QEMU capture shows that `cd d` succeeds at a prompt, a later unqualified `ls` fails with `exec ls failed`, and a relative executable path works (Figure 2.3).

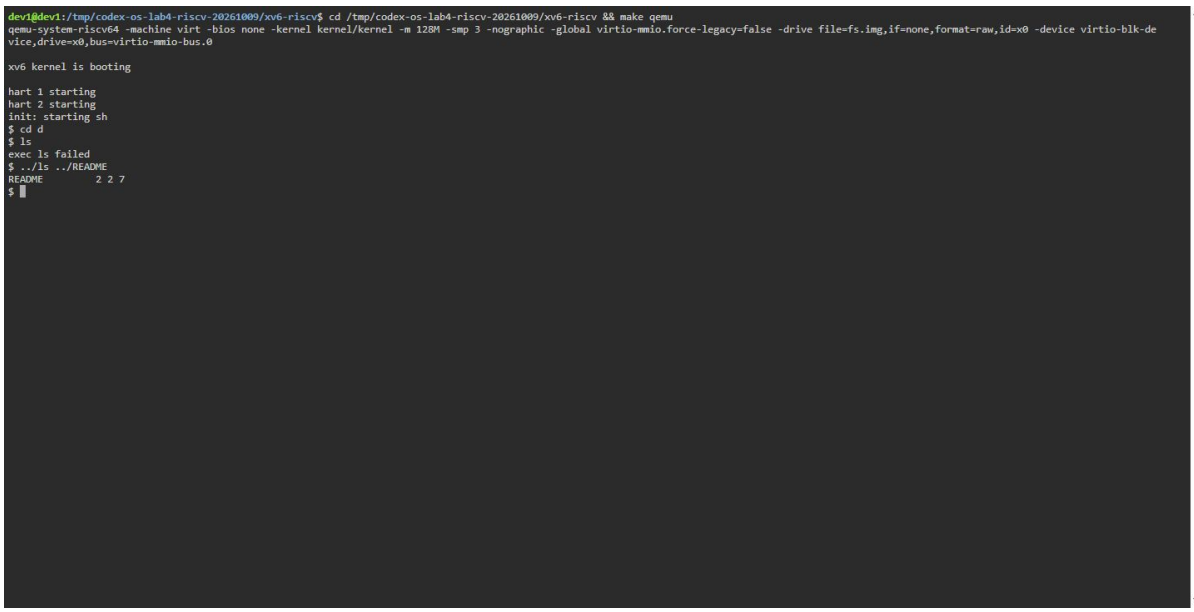


Figure 2.3 – Terminal capture, Dev1 Debian GNU/Linux 13, 10 October 2026: path lookup after changing directories in xv6.

2.4 Answers to Part 2 Observe questions

a) **List the features both shells have, and the ones only Bash has.**

Features in both: Direct command execution, file descriptor redirection (<, >), pipelines (|), parenthesized command grouping, sequential execution (;), and asynchronous background execution (&). The grouped xv6 command in the test returned `2 2 4`.

Features only in Bash: Globbing (*, ?, `[...]`), environment variables (`$VAR`), quote processing (`"..."`, `'...'`), conditional operators (&&, ||), automatic `PATH` search, history search, and tab completion;

b) **In xv6, `ls` stops working after `cd d`. Why? Find a way to run `ls` from inside `d` anyway.**

At the top-level prompt, `cd d` succeeds. The xv6 shell has no `PATH` lookup, so `ls` is resolved relative to `d` and fails. Figure 2.3 shows the working relative path `../ls ../README`, which explicitly invokes the executable in the parent directory;

c) **Which missing feature would you miss most in daily work, and which would be easiest to add?**

Most missed: `PATH` directory resolution and tab completion/history. Without `PATH`, changing directories renders all core utilities inaccessible unless typed with full paths.

Easiest to add: A fallback path search in `user/sh.c`. If `exec(cmd, argv)` returns failure, prepend `"/"` or `"/bin/"` to `cmd` and re-execute before returning an error to the user;

3 ARGUMENT TOKENIZATION: WHO EXPANDS WILDCARDS?

3.1 Implementation of argument vector inspectors

To compare how Bash and xv6 prepare arguments, identical inspection utilities were authored for xv6 (`user/args.c`) and Linux (`args_linux.c`). Each prints `argc` and each `argv[i]` string enclosed in brackets. Figure 3.1 illustrates this boundary at a high level; it does not specify Bash's complete expansion-order algorithm.

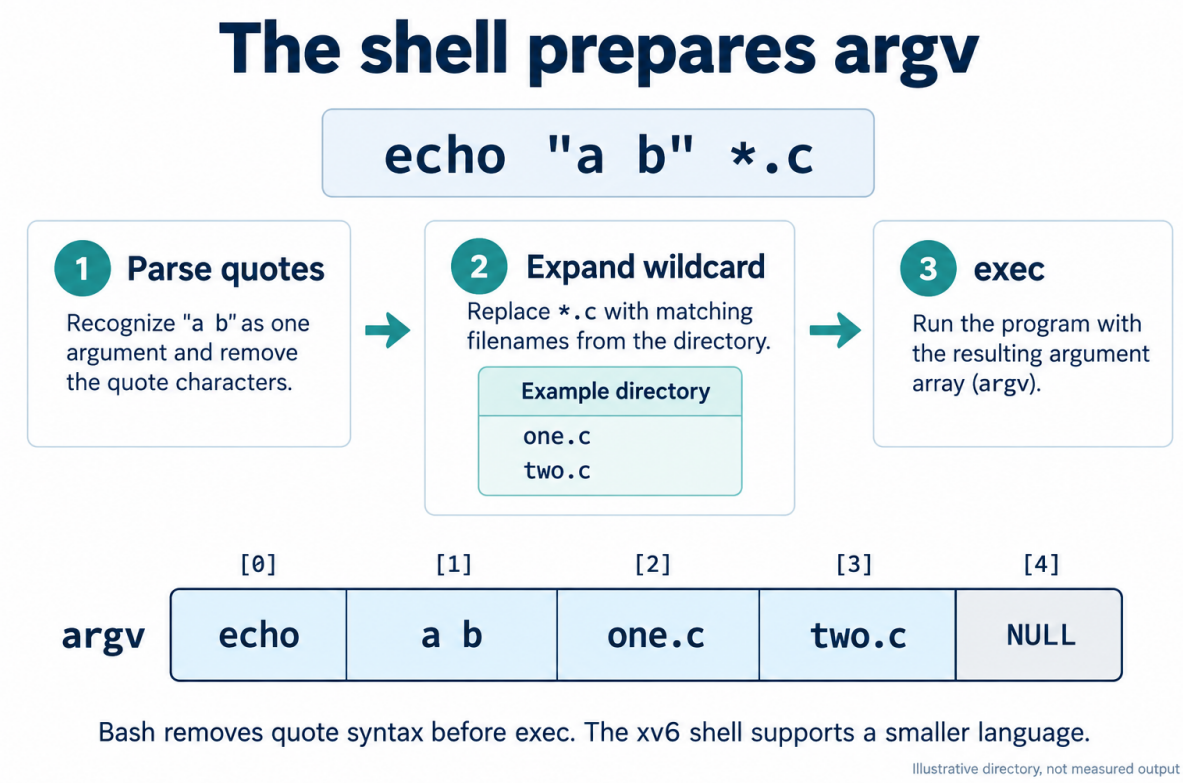


Figure 3.1 – Conceptual illustration of shell argument preparation before program execution; not an execution capture or a complete Bash expansion-order specification.

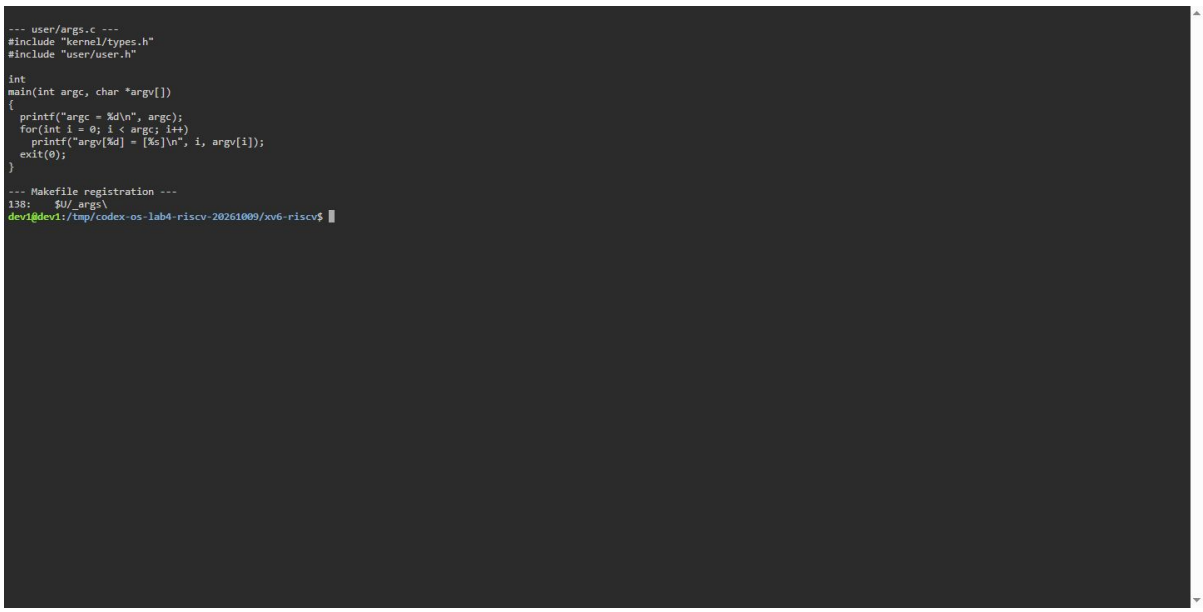


Figure 3.2 – Terminal capture, Dev1 Debian GNU/Linux 13, 10 October 2026: Source of user/args.c and registration within xv6 Makefile UPROGS.

The copied `Makefile` registers both `$U/_sleep` and `$U/_args` in `UPROGS`, so the build includes the utilities in `fs.img`. The capture shows the source and registration lines; the 9 and 10 October QEMU runs built and booted this temporary tree.

3.2 Experimental comparison across shells

Both implementations were run with all four requested argument patterns. The Linux capture used a working directory containing `a.c`, `b.c`, and `args_linux.c`; the xv6 capture shows the arguments received in QEMU:

1. `args one two three;`
2. `args *.c "x y";`
3. `args $HOME;`
4. `args 'single quoted';`

Figure 3.3 shows the actual output from all four xv6 invocations.

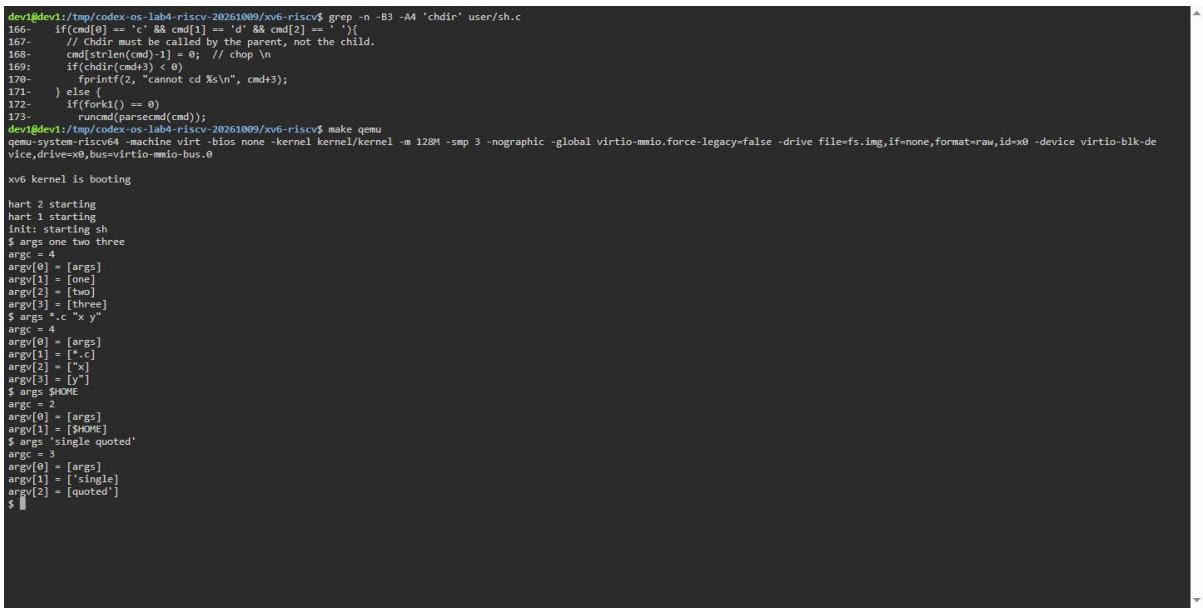


Figure 3.3 – Terminal capture, Dev1 Debian GNU/Linux 13, 10 October 2026: xv6 output for all four `args` patterns.

Figure 3.4 shows the actual Linux Bash output from the same four invocations.

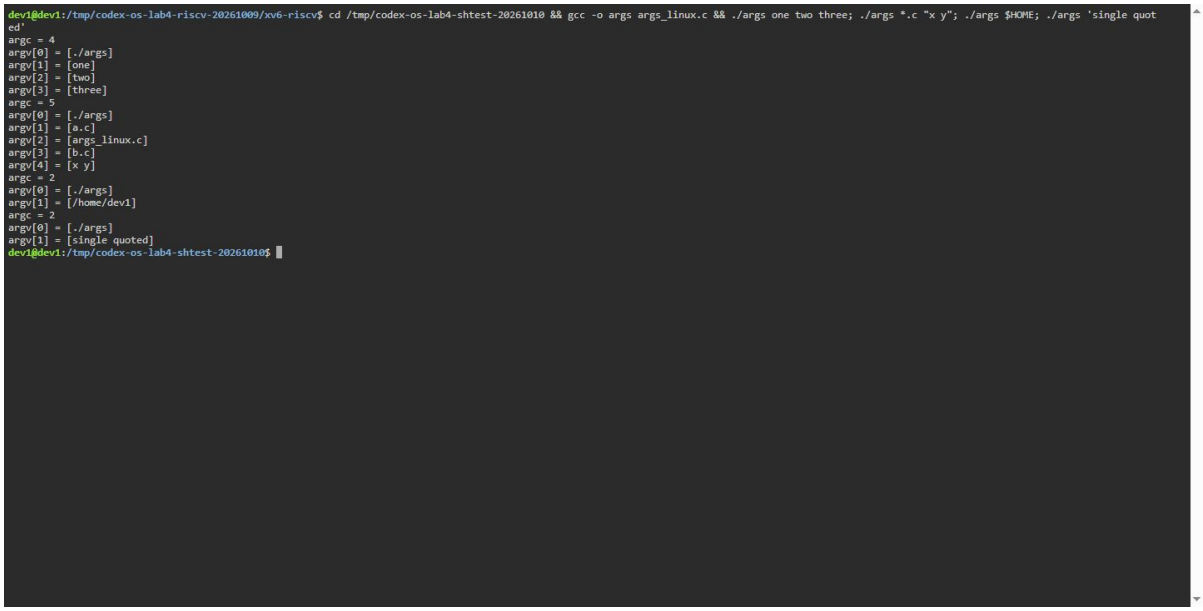


Figure 3.4 – Terminal capture, Dev1 Debian GNU/Linux 13, 10 October 2026: Linux Bash output for all four `args` patterns.

3.3 Answers to Part 3 Observe questions

a) **Paste both outputs for `args *.c "x y"`. How many arguments did each program receive?**

In Linux Bash:

```
argc = 5
argv[0] = [./args]
argv[1] = [a.c]
```

```
argv[2] = [args_linux.c]
argv[3] = [b.c]
argv[4] = [x y]
```

In xv6 Shell:

```
argc = 4
argv[0] = [args]
argv[1] = [*.c]
argv[2] = ["x]
argv[3] = [y"]
```

The Linux program received 5 arguments because the wildcard matched three source files and the quoted phrase remained one argument. xv6 received 4 arguments: it preserved `*.c` literally and split ẍ ÿ into two strings with quotes intact;

b) **Who turns `*.c` into file names and removes quotes: the program, the kernel, or the shell?**
The shell. The shell performs lexical scanning, globbing, variable expansion, and quote stripping before invoking `exec()`. The kernel merely accepts an array of already-expanded null-terminated strings and copies them to the new process stack. The user program receives the strings exactly as prepared by the shell;

c) **What happened to the extra spaces in `one two three`?**
The shell treats spaces as argument separators (delimiters). Multiple contiguous spaces are consumed by the lexer during tokenization, so only the non-whitespace tokens (`"one"`, `"two"`, `"three"`) are inserted into the argument array;

4 ARCHITECTURE OF USER/SH.C

4.1 Command representations and syntax nodes

The xv6 shell represents commands using tagged C structures. Five command types are defined on lines 8–12 of `user/sh.c` (Figure 4.1):

- `EXEC (1)`: Basic command execution; no operator;
- `REDIR (2)`: Redirection of I/O; produced by < and >;
- `PIPE (3)`: Inter-process communication; produced by `|`;
- `LIST (4)`: Sequential commands; produced by `;`;
- `BACK (5)`: Asynchronous background execution; produced by `&`;

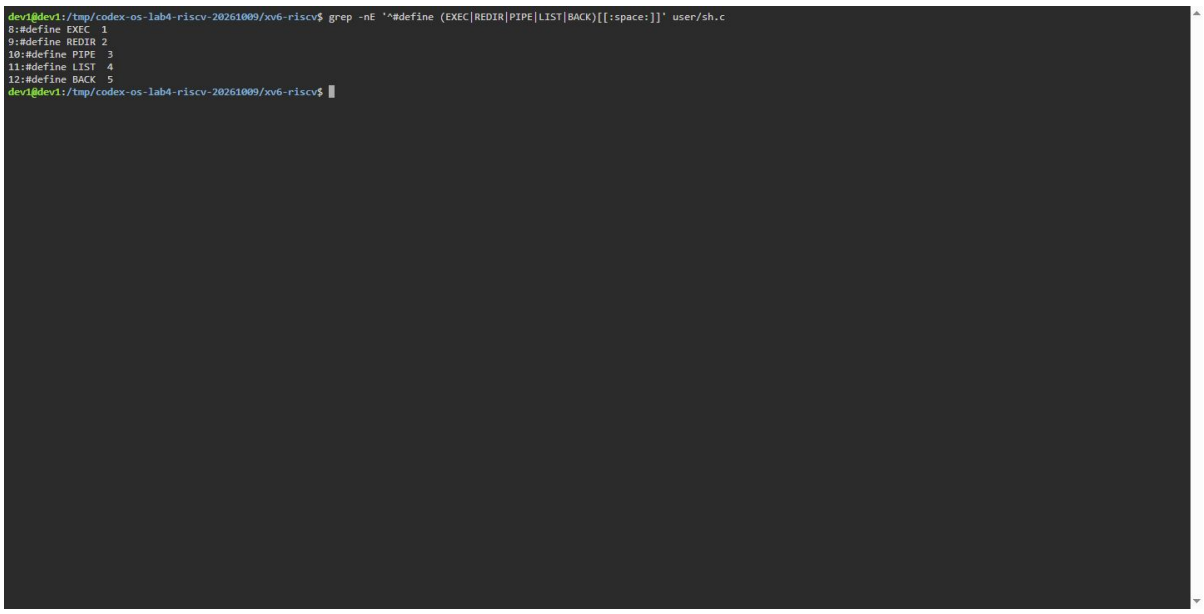


Figure 4.1 – Terminal capture, Dev1 Debian GNU/Linux 13, 10 October 2026: Command types defined in user/sh.c.

4.2 Recursive command execution in runcmd()

When a command line is parsed, `parsecmd()` constructs an abstract syntax tree composed of these structures. The child process then invokes `runcmd()` [1] (Figure 4.2).

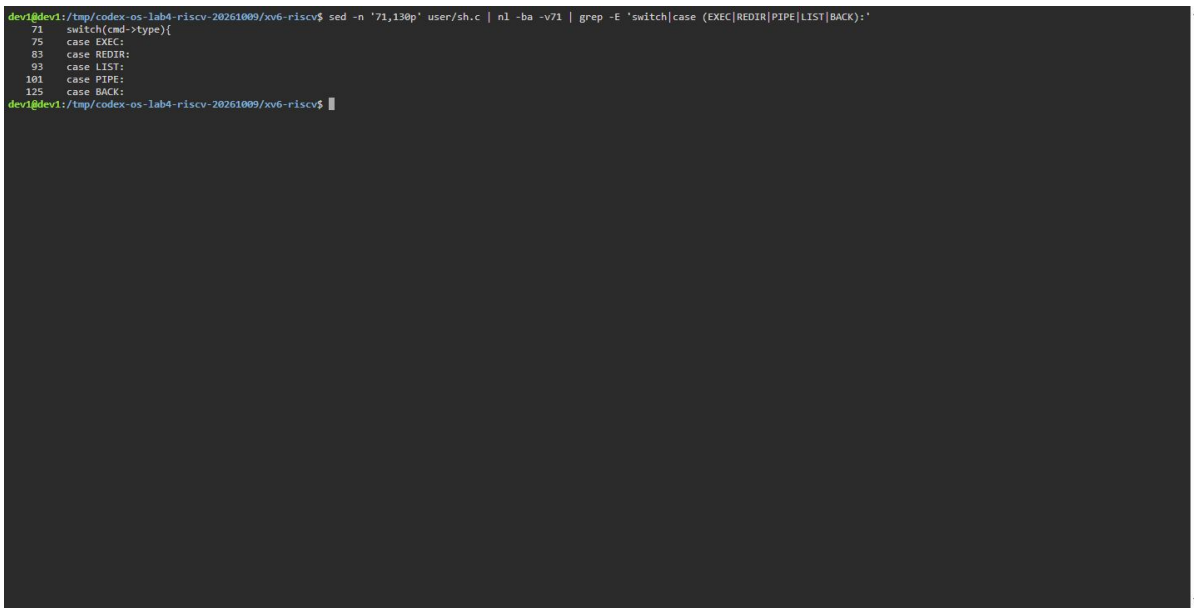


Figure 4.2 – Terminal capture, Dev1 Debian GNU/Linux 13, 10 October 2026: Dispatch switch inside runcmd() handling syntax tree execution.

The execution logic handles each node type:

- For `EXEC`, it invokes `exec(ecmd->argv[0], ecmd->argv);`
- For `REDIR`, it closes the targeted descriptor, opens the named file with the requested flags, and recursively evaluates the inner command;
- For `PIPE`, it calls `pipe()`, forks two sub-processes, rewires descriptors 0 and 1 using `dup()`, closes pipe ends, and waits for both children to terminate;
- For `LIST`, it forks a child to run the left command, waits for it, and then runs the right command;
- For `BACK`, it forks a child to run the command and immediately returns without waiting;

4.3 Built-in cd handling and process boundaries

Lines 166–170 of `user/sh.c` reveal that `cd` is explicitly intercepted in the main shell loop before any call to `fork()` (Figure 4.3).

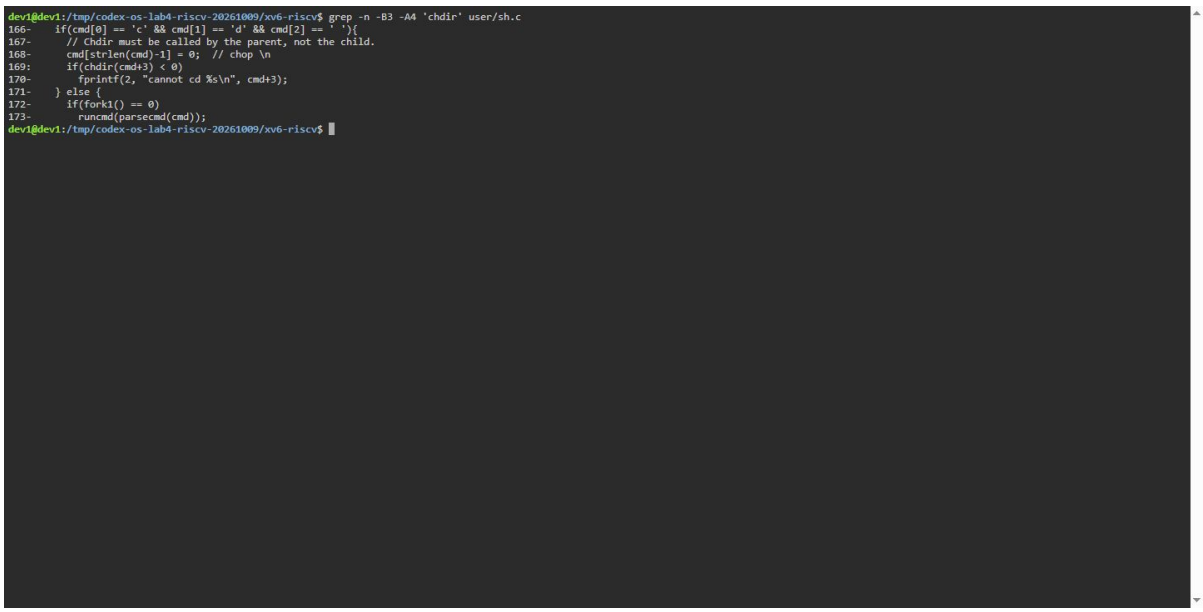


Figure 4.3 – Terminal capture, Dev1 Debian GNU/Linux 13, 10 October 2026: Parent-process handling of cd built-in in user/sh.c.

4.4 Answers to Part 4 Observe questions

a) **Name the five kinds of command the xv6 shell knows and the character that produces each.**
`EXEC` (basic command, no special character), `REDIR` (produced by < and >), `PIPE` (produced by `|`), `LIST` (produced by `;`), and `BACK` (produced by `&`);

b) **Where is `cd` handled, and why is it not run as a separate program?**
`cd` is handled in `main()` of `user/sh.c` prior to the `fork()` call. It cannot be run as a separate program because each child process operates in its own isolated address space and process descriptor table. If `cd` were an external program, only the child process would change its directory; upon child exit, the parent shell would remain in the original directory. Therefore, `chdir()` must execute directly in the parent shell process;

c) **Search `sh.c` for any code that handles `*` or `$`. What does that explain about Part 2?**
Searching `sh.c` reveals zero code handling `*` or `$`. Neither globbing nor variable expansion logic exists in the parser. This explains why `echo $HOME` and `ls *` print literal characters in xv6: the shell treats them as ordinary alphanumeric tokens and passes them uninspected to `exec()`;

CONCLUSIONS

The captures distinguish Bash expansion from xv6 token splitting, while the source figures identify the xv6 command types and parent-process handling of `cd`. A focused QEMU check found no Up-arrow recall or Ctrl+R search, and Tab at an empty prompt produced no completion list. Source inspection confirms that the xv6 console has no history, completion, or reverse-search handler; path-prefix completion was not tested.

REFERENCES

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2. Love, R. (2013). *Linux System Programming: Talking Directly to the Kernel and C Library* (2nd ed.). O'Reilly Media.

3. Bach, M. J. (1986). *The Design of the UNIX Operating System*. Prentice-Hall.